WildGuard

Cross-Dataset Label Transfer for Wildlife-Trade Risk Prioritization

GROUP 37 | PRS PRACTICE MODULE

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Problem

Wildlife trade happens at a scale no enforcement agency can manually inspect, and the risk signal lives in patterns across fields rather than any single one.

Scale beyond manual review

CITES holds more than 25 million trade records dating back to 1975, far beyond what any human reviewer could assess one by one.

The signal is a pattern, not a field

Risk isn't visible in any single field. It's a pattern across species, quantity, route, purpose and value, relative to historical norms, and that's exactly what pattern recognition is built to find.

A recognised gap, not yet tooled

The CITES Secretariat and WCO confirmed in December 2025 that agencies already want risk-based targeting, but today it's done manually, one officer at a time. That's exactly the kind of scale mismatch pattern recognition is built to close.

Changes to Project Direction & Scope

Continue with structured + visual fusion (pangolin images)?

Branch A: Modality / Fairness

- Visual = pangolins only; structured = all species
- Missing visual data would skew scores unfairly right now with the current approach
- "Hybrid beats single-stream" is unprovable for one species, let alone all species or cross-species
- Streams may not even describe the same shipment

Branch B: Usefulness

- Would only show what's already known (species-level risk)
- Not case-specific urgency a human didn't already know

Branch C: Data Feasibility

- Synthetic augmentation leads to a synthetic-to-real domain gap
- Cardoso et al. dataset request: no response

DECISION: Drop the visual + structured fusion design

New direction: CITES and LEMIS, using single-modality, cross-dataset label transfer

Project Objectives

How can real enforcement outcomes recorded in one national wildlife-trade dataset (LEMIS) be used to build a statistically-grounded risk-scoring system that prioritises shipments for human review within the much larger, unlabeled global CITES trade database, closing the gap between the officially-recognised need for risk-based targeting (CITES Secretariat/WCO, Dec 2025) and the tools to actually do it at CITES's real data scale?

1. Acquire & align schemas

LEMIS + CITES field/code mapping, verified: source/purpose codes match almost exactly.

2. Engineer risk features

Price-per-unit, quantity deviation, route frequency, purpose/source context.

3. Train supervised classifier

On LEMIS's real enforcement labels, using an ensemble method (RF / XGBoost).

4. Run unsupervised detection

On CITES's unlabeled stream, using clustering / statistical outlier detection.

5. Combine LEMIS-trained score + CITES anomaly score → one hybrid review-priority output

6. Validate: two-part plan (see Evaluation Plan)

7. MVP: interface showing flagged shipments, priority tier, and explanation

Data Source Decision

CITES has the scale but no labels. LEMIS has the labels but no scale. Verified schema compatibility so that they can be combined → the application gap that has not been filled yet.

CITES Trade Database

Gives: *scale*

- 25M+ shipment records since 1975
- Public, bulk-downloadable (trade.cites.org)
- Every CITES-listed species, every reporting country

Lacks: *reliable enforcement / fraud labels of its own.*

LEMIS (US F&W Service)

Gives: *labels*

- ~1M shipments / 5.5M+ line items, 2000–2014
- Free via Zenodo (CC-BY-4.0), no registration
- Real outcomes: Cleared / Refused / Seized / Abandoned

Lacks: *US-only, with no global scale on its own.*

How did we verify schema compatibility?

Checked that LEMIS's source/purpose codes match CITES's coding almost exactly

(e.g. "I" = Confiscated/seized specimen in both), since LEMIS was built to log CITES-relevant declarations.

Technical Approach: System Design

1. Schema Alignment

KNOWLEDGE MODELLING

Map LEMIS ↔ CITES fields and codes.

EXAMPLE OUTPUT

source=W (LEMIS) ↔ source=W (CITES) = "wild-sourced"

2. Supervised Classifier

SUPERVISED · RF / XGBOOST

Trained on LEMIS's real enforcement labels.

EXAMPLE OUTPUT

P(Seized/Abandoned) = 0.82 for a given record profile

3. Anomaly Detection

UNSUPERVISED

Run on CITES's unlabeled shipment stream.

EXAMPLE OUTPUT

Tortoise shipment: $400/unit vs. $6 peer median (≈67x)

4. Score Combination

HYBRID / ENSEMBLE

Combine LEMIS-trained + CITES anomaly scores.

EXAMPLE OUTPUT

Combined priority: HIGH

5. Human Review

OUTPUT LAYER

Flagged shipment card with explanation.

EXAMPLE OUTPUT

Geochelone elegans, 2 live, wild, Personal, MY→US (actually Seized)

Evaluation Plan

Primary: LEMIS (labelled ground truth)

Held-out temporal split (train earlier years, test later).
Compare supervised-only vs. unsupervised-only (on LEMIS's own price/quantity/route fields) vs. hybrid, using precision, recall, F1, PR-AUC, not accuracy.
With a real LEMIS sample, found non-Cleared outcomes (Seized/Abandoned/Reexport) were only ~2–3% of records, hence PR-AUC/F1 over accuracy.

Secondary: CITES (no ground truth)

No exact fraud-case matching (checked: neither candidate source gives matchable shipment records).
Instead: (a) qualitative top-N review of flagged shipments for domain plausibility, and (b) an aggregate pattern-consistency check, asking whether flagged shipments disproportionately involve species/routes/countries already flagged as high-risk in aggregate seizure statistics (UNODC WWCR 2024, TRAFFIC).
Plausibility and directional consistency only, not a precision/recall claim.

Also: *cluster stability & silhouette scores as a baseline sanity check for the unsupervised component (not the headline result).*

Caveat:

LEMIS reflects US enforcement patterns; applying it to CITES's global trade crosses a real distribution shift: a genuine limitation, stated plainly, not hidden.

What We Have Done So Far

This presentation is about a revised plan, not a built system yet. Here is the honest state of progress.

DONE

- Verified both datasets are real, large, and immediately accessible (no request/waiting period), de-risking the single biggest failure mode of the original design.
- Verified schema compatibility between LEMIS and CITES at the code level (not assumed).
- Researched and confirmed that this specific cross-dataset approach hasn't been published before; found the closest real-world research evidence for why it matters (CITES/WCO Dec 2025 guidance).
- Pulled and validated a real worked example from actual LEMIS data to confirm the kind of signal the approach would pick up on.

NOT YET DONE

- Actual model training and evaluation
- Full CITES bulk download + schema-alignment implementation
- MVP build